

RESEARCH ARTICLE

Intention Recognition of Space Noncooperative Targets Using Large Language Models

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This study proposes a novel method for intention recognition of space noncooperative targets using large language models (LLMs). Traditional methods rely on motion data to assess orbital motion intentions but cannot infer operation and task intentions from multi-source information like images. LLMs, with their logical reasoning capabilities, can address this limitation. The intentions are categorized into 3 types and 23 subtypes based on multi-source information and their characteristics: motion intentions (e.g., “hovering”, “flyby”, and “rendezvous”), operation intentions (e.g., “docking”, “refueling”, and “repair”), and task intentions (e.g., “detection”, “surveillance”, and “attack”). The proposed method constructs LLMs for spacecraft intention recognition, involving prompt classification, template design, and test sample generation. The use of prompt tuning V2 (P-tuning V2) and low-rank adaptation (LoRA) fine-tuning enhances the models' performance. A dataset of 50,688 nominal samples and 8,448 perturbed samples was created through computer simulation based on expert knowledge, focusing on intention recognition of approaching targets in space station on-orbit operation and surveillance scenarios. The models were tested under 3 prompt conditions: basic, instruction, and chain-of-thought (CoT). The performance of 6 models (ChatGLM2-6B and ChatGLM3-6B base and fine-tuned models) was analyzed. Notably, the LoRA fine-tuned ChatGLM3-6B model on instruction prompts achieved 99.9% accuracy, with improved robustness compared to the base model. This work presents a pioneering application of LLMs for spacecraft intention recognition, offering valuable insights for future research and applications.

Introduction

As the number of artificial satellites in Earth's orbit soars, space security confronts unprecedented challenges [1,2]. Concurrently, noncooperative space targets, including abandoned satellites, space debris, and unidentified satellites, pose a grave threat to the safety of space activities [3]. The increasing threats to satellites in orbit has necessitated the implementation of orbit safety warning [4]. Traditional on-orbit threat warning mechanisms primarily rely on assessing the minimum distance between satellites [5], which substantially limits their effectiveness [6]. These systems fail to adequately account for the relative orbital motion characteristics between operational spacecraft and noncooperative targets [7]. Additionally, they do not consider the potential intentions, maneuvering patterns, and specific characteristics of these noncooperative targets, which can lead to false reports and compromised situational awareness. To enhance the efficacy of early warning systems, there is an urgent need to develop more comprehensive threat assessment method. The method must integrate in-depth analysis of target behavior, understanding of relative motion, identification of target types, and assessment of potential impacts on orbital assets and missions [8]. Spacecraft intention recognition enables operational spacecraft to make informed decisions by swiftly identifying the intentions

of noncooperative targets in space, thereby addressing the aforementioned issues.

With the exponential growth of computer technology, the recognition of intentions has emerged as a pivotal area of research within the realms of artificial intelligence and machine learning. Dating back to 1986, Kautz and Allen [9] pioneered a theory of plan recognition that encapsulates the vast expanse of intention recognition research to this day. Over time, 3 principal research methodologies have crystallized for intention recognition: statistical methods, supervised learning, and deep learning. Among the prevalent statistical methods are support vector machines (SVMs) [10], naive Bayes classifiers [11], and others. These methods stand out for their robust interpretability, high computational efficiency, and suitability for small datasets; however, they are hindered by limited generalization capabilities, challenges in managing high-dimensional data, and a susceptibility to noise. Supervised learning, on the other hand, introduces a wealth of labeled training data, incorporating classification algorithms such as decision trees [12], maximum entropy models, and more. These methods offer strong interpretability, robust generalization capabilities, and ease of evaluation but are contingent upon extensive labeled data, may grapple with overfitting issues, and face difficulties in model updates. Deep learning approaches leverage complex networks like recurrent neural networks (RNNs) [13],

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long short-term memory networks (LSTMs) [14], and others. These methods are distinguished by their higher accuracy, the ability to handle large-scale data, strong feature learning capabilities, and robust generalization abilities. However, they demand large amounts of training data, significant computational resources, and extended training periods and lack model interpretability. While these methods are the bedrock of intention recognition, a holistic intention recognition concept and methodology tailored for spacecraft have not yet been fully realized [15].

Nevertheless, scholars have commenced addressing these challenges. Yang et al. [16] have introduced 3 indicators—"fly-by", "hit", and "angle"—as key criteria for space situational assessment. By employing cluster analysis and membership division, they have successfully fuzzified the characteristic data of target intentions and utilized a fuzzy decision tree algorithm to establish a pioneering intention recognition model. Qi et al. [17] have pioneered a novel method and system for determining the intentions of noncooperative target spacecraft. Their innovation categorizes these intentions into 5 categories: orbit maintenance, approach, evasion, interception, and accompany flight. By manually annotating orbital dynamics data, they trained an RNN, resulting in a robust network capable of identifying the intentions of noncooperative target spacecraft.

A landmark study in this arena is the work conducted by Sun and Dang [18], which broke new ground by developing a model for identifying the intentions of noncooperative targets through the prism of deep learning theory and algorithms. The research team commenced by clarifying the critical concept of "spacecraft intention" and constructed a mathematical model for "intention recognition". Anchored in the relative orbital dynamics model Clohessy–Wiltshire (CW) equation, they outlined 10 discrete types of noncooperative target motion intentions. Harnessing the power of deep learning technology, they engineered a model skilled in identifying such intentions. To ensure widespread utility, the study utilized satellite orbit extrapolation to generate the necessary dataset, which was subsequently employed to train backpropagation deep neural networks. The result of stringent training and testing was an impressive recognition accuracy rate of 92.89%. This model, rooted in deep neural networks, outperformed its counterparts in terms of effectiveness, adaptability, and computational efficiency. Yet, the research is not without its constraints; it is capable of discerning only specific, predefined intentions, thus falling short of comprehensiveness. Furthermore, the information it relies on for intention recognition is limited to relative orbital positions and velocities, without incorporating additional contextual factors (such as spacecraft geometry and functional analysis), which could potentially enhance accuracy.

There are 2 significant deficiencies in the current research, although the above authors made significant progress on spacecraft intention recognition. First, there is a lack of a well-defined and categorized framework for understanding spacecraft intention. Second, the spacecraft intention recognition is predominantly confined to the analysis of observed orbital motion data, without fully leveraging additional information such as the target's shape or environmental factors, which can lead to erroneous judgments and have a substantial impact on subsequent mission.

To address the first deficiency, in the context of safety warning tasks during the operation of a space station, this paper systematically explores the definition and conceptual connotations of

the intentions of incoming noncooperative space targets from the perspective of the space station. Drawing on the characteristics of intention vocabulary, the paper categorizes spacecraft intention and conducts an in-depth analysis of the specific intention terms under each category, ultimately establishing a linguistic model of spacecraft intention that encompasses 3 categories and 23 types of intention vocabularies. Although this classification may not be exhaustive, it is adequate for assessing the capability of large language models (LLMs) in recognizing spacecraft intentions. To extract prompts for specific intention relevant to space station maintenance and operations, the paper distills the scene information into 4 categories and 7 types of prompt elements for spacecraft intention recognition tasks. To facilitate interaction with LLMs, a standardized template for input prompts is developed based on these 4 categories of hint elements and directive prompt statements. By incorporating these prompt elements into the standard template, prompt statements are generated. The integration of prompt statements with corresponding spacecraft intention generates the LLM test samples.

To rectify the second deficiency, the present study converts diverse data sources—such as orbital trajectory data, imagery and video data, and environmental context data—into a corpus of multi-source textual information for ingestion by LLMs. This approach leverages the LLMs' capacity for synthetic reasoning and evaluation of textual content. In pursuit of this objective, the research delves into an exploration of the inferential capabilities inherent in LLMs. Compared to previous research, Wei et al. [19] have significantly enhanced the reasoning ability of LLMs through the chain-of-thought (CoT) prompt method. The core principle of CoT prompt is to encourage LLMs to elucidate their reasoning processes. The CoT implementation involves presenting LLMs with a limited set of examples that include explained reasoning processes. When responding to prompts, LLMs' output also reveal detailed reasoning processes. The research findings indicate that as model parameters increase, LLMs exhibit a progressive improvement in answer accuracy. In general, the answer accuracy of the CoT prompt model exceeds that of the standard prompt model, and in certain domains, such as sports, the CoT prompt method exhibits higher accuracy in answering questions than humans. By synthesizing the CoT approach, the general structure of CoT is delineated. A comprehensive CoT prompt typically comprises 3 elements: instruction, rationale, and exemplars. Instructions generally serve to articulate the problem and specify the desired output format for the large model. Rationale refers to the intermediate reasoning process of CoT, which may include solutions to the problem, intermediate inference steps, and any relevant external knowledge. Exemplars, on the other hand, provide the basic input–output pair format for the large model in a few-shot manner, with each exemplar consisting of a question, the reasoning process, and the corresponding answer.

Following CoT, Wang et al. [20] have introduced the self-consistency method, which primarily enhances answer accuracy through majority voting and markedly improves the performance of the CoT method. Zhou et al. [21] have applied the least to most prompting (LtM) method, which builds on the CoT prompt process by initially breaking down a problem into subproblems and then addressing them sequentially. Unlike CoT, LtM incorporates the solutions to previous subproblems into the prompts to attempt to solve subsequent problems. Simply put, while pure CoT may not suffice for complex

problems, breaking them into smaller issues and then applying CoT enables the model to resolve them. Overall, the advent of CoT has significantly enhanced the logical reasoning abilities of LLMs, making them suitable for various information inputs and spacecraft intention recognition domains that involve complex logical reasoning. However, there is no universally accepted theory that explains why CoT improves a model's reasoning ability. Beyond this, a multitude of papers have conducted experiments on the interaction between CoT and large models, showing that CoT markedly intensifies the logical reasoning abilities of LLMs. Empirical evidence from extensive experiments suggests that the application of CoT prompts in LLMs adapts them for various information inputs and the field of spacecraft intent recognition involving complex logical reasoning.

The rest of the paper is organized as follows. In Materials and Methods, this paper outlines the construction of a language model for recognizing spacecraft intentions. This includes defining and classifying spacecraft intentions, categorizing prompt elements for interacting with LLMs, designing prompt templates, and developing test samples for evaluating LLMs. Additionally, it introduces 2 fine-tuning methods to enhance LLM performance: prompt tuning V2 (P-tuning V2) and low-rank adaptation (LoRA) fine-tuning. The Results section presents the performance of 6 models—ChatGLM2-6B and ChatGLM3-6B base models and their fine-tuned versions—under 3 different prompt conditions, along with robustness tests conducted across 3 scenarios. The ChatGLM3-6B model, fine-tuned with instruction prompt data using LoRA, achieved an accuracy of 99.90% with robustness close to that of the base model. Similarly, the ChatGLM2-6B model fine-tuned the CoT prompt data using P-tuning and achieved an accuracy of 99.81% in testing, demonstrating the best robustness performance. These results indicate the powerful capability of LLMs in accurately interpreting the intentions of noncooperative space targets.

Materials and Methods

Scenario description

To enhance understanding of the article's objective, we first describe the application scenarios of spacecraft intention recognition. As shown in Fig. 1, operational spacecraft (**O**) observes target spacecraft (**T**) at time t_0 . The predicted trajectory of **T** is depicted by the blue dashed line; however, after a period of observation, the actual trajectory of **T** is represented by the black solid line. **O**, through its camera, identifies that **T**

is equipped with some devices such as a robotic arm. At this moment, the surrounding lighting is favorable. For its own safety and subsequent decision-making, **O** must consider the intention of **T**. The aim of this article is to convert these multi-source information into input text for LLMs, enabling them to ascertain the potential intentions of **T**.

Design and construction of LLM-based intention recognition method

Essentially, the study of textual language follows a hierarchical structure: starting with a collection of documents, then individual documents, paragraphs, sentences, phrases, words, and finally characters [22]. To construct text that interacts with LLMs, we follow the reverse order of the research hierarchy mentioned above, constructing words first, then sentences, and finally paragraphs.

The LLM can be represented as $P_{\Phi_0}(Y|X)$, where X is the input text, Y is the output to be predicted, and Φ_0 represents the parameters of the LLM. To employ an LLM for spacecraft intention recognition, the initial step is to articulate the spacecraft intention in textual (i.e., construct the expected output Y_e) form to produce accurate labels that the model can output. Additionally, to facilitate interaction with LLMs, it is essential to design prompt words and templates specifically for LLMs (i.e., construct the input X). Finally, to verify the output accuracy of LLMs for intention recognition, it is necessary to design a nominal “input information X -expected output intentions Y_e ” validation dataset.

Representation of spacecraft intentions

In unraveling the conceptual essence of spacecraft intentions, we must delineate its meaning from the basics of human prior knowledge and cognition. The Cambridge Dictionary defines “intention” as “something that you want and plan to do”. From a definitional standpoint and in common sense, the subject of intention is always a human or an intelligent agent. The objective of intention recognition is to deduce the subject's intent by observing actions.

In addressing the problem of spacecraft intention recognition, our initial assumption is that the spacecraft being identified has performed some form of orbital maneuver (action), and this maneuver is driven by a certain intention [23]. It may originate from manipulating the intention of operator. Additionally, research does not encompass complex scenarios where spacecraft behavior may not be driven by distinct intentions. From the aforementioned studies, we define the lexical nature of spacecraft intention as the action performed by the target spacecraft to achieve a particular purpose, with the lexical nature being either a verb or a noun describing the action process.

In essence, executing specific actions to achieve a predetermined purpose can be seen as the manifestation of intent. In common understanding, we can assume that these purposes are the intents that need to be identified. For example, in the scenario where the target spacecraft is designed to capture an operational spacecraft, the intentions “grabbing” or “grasping” must be recognized. Words like “grabbing” inherently possess the attribute of action (operation). Similarly, if the target spacecraft is intended to collide operational spacecraft, words like “collision” or “crash” possess the attribute of action but lack specific operational attributes, more akin to task-level attributes.

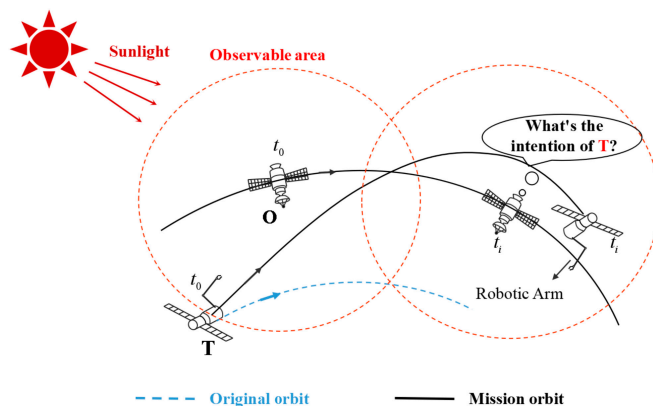


Fig. 1. Typical scenario in spacecraft intention recognition.

Previous work conducted preliminary research on spacecraft intention recognition under a narrow set of conditions, representing the spacecraft's intentions as patterns of relative orbital motion [17,18]. It appears that spacecraft intention words themselves possess different attributes and can be categorized into different types. To delve deeper into the essence of spacecraft intention and ensure the systematic, accurate, and practical nature of our research, we categorize spacecraft intentions into 3 attributes: motion intentions, operation intentions, and task intentions.

Several experts discussed and summarized the possible intentions, expanded the 3 types of spacecraft intention attributes mentioned above, and formed a spacecraft intention corpus, as shown in Table 1. However, the content may have omissions. Each intention in Table 1 has specific connotations. We test the use of Chinese as the input and output language for the LLMs. The subsequent prompt elements, standard prompt statements, test samples, LLM input examples, and LLMs output results are all in Chinese, and this paper translates them into English for better understanding.

A schematic representation of typical intention images is presented, as depicted in Fig. 2. The motion intention is represented by a black solid line, indicating the target's trajectory in relation to the operating spacecraft. The operation intention highlights the potential devices carried by the target spacecraft, including mechanical arms, cameras, fuel package, communication antennas, and scientific experimental equipment. The task intention is a combination of both motion and operation intentions, thus illustrating both the relative motion trajectory via the black solid line and the devices aboard the target spacecraft. Regarding environmental conditions, only the lighting conditions are visually represented, with no depiction of electromagnetic conditions.

Therefore, we defined the expected outputs of Y_e LLMs as:

$$Y_e = \{y_1, y_2, y_3\} \quad (1)$$

Here, y_1 represents the motion intentions, y_2 represents the operation intentions, and y_3 represents the task intentions, where

$$\begin{aligned} y_1 &\in \{\text{"hovering"}, \text{"flyby"}, \text{"flyaround"}, \text{"rendezvous"}, \text{"collision"}, \text{"retreat"}, \text{"randomness"}\} \\ y_2 &\in \{\text{"docking"}, \text{"refueling"}, \text{"repair"}, \text{"grabbing"}, \text{"photograph"}, \text{"communication"}\} \\ y_3 &\in \{\text{"detection"}, \text{"surveillance"}, \text{"interference"}, \text{"negotiation"}, \text{"observation"}, \text{"assistance"}, \text{"deterrence"}, \text{"support"}, \text{"experiment"}, \text{"attack"}\} \end{aligned} \quad (2)$$

It is mandatory for all expected outputs to include y_1 , whereas the presence of the other 2 intentions, y_2 and y_3 , is not always required.

Design of prompt elements for interaction with LLMs

The prompt elements X_p that we constructed are a part of the input X . The design prompt elements fundamentally extract scene information for spacecraft intention recognition and translates it into text input for LLMs. Mimicking the human thought process, inferring spacecraft intentions requires a given scenario from which information is extracted to determine possible intentions. Prompt elements serve as the input information distilled from the scenario for LLMs to infer. For instance, if the intention is "repair", the corresponding elements might include the malfunction of operational spacecraft,

the orbital rendezvous with the target spacecraft, the target carrying repair tools, and the target being a cooperative entity, among others.

When selecting prompt elements, the information obtained from actual sensors must be considered. To meet the requirements of LLMs, this information must be converted into textual form. This process disregards the temporal order of information acquisition, instead assuming that all information can be gathered and transformed into prompt elements, with the sequence of these elements determined subsequently. Spacecraft orbital missions are influenced by numerous factors, such as electromagnetic conditions and Earth's shadow regions. In our study, we pay special attention to these factors to more accurately infer the specific intentions of spacecraft.

To simplify the inference of intentions in typical scenarios, we chose the threat warning tasks during the operation of the space station as the basic scenario for this paper. We categorize and summarize the information needed to infer intentions into prompt elements, specifically including information about operational spacecraft's information x_{p1} , target information x_{p2} , environmental conditions x_{p3} , and orbital motion characteristics x_{p4} , where

$$X_p = \{x_{p1}, x_{p2}, x_{p3}, x_{p4}\} \quad (3)$$

The identity information for operational spacecraft includes its function, its Earth orbit, and its inherent value. (In the experiment, to simplify the intention recognition scenario, we chose operational spacecraft as a space station in a low Earth orbit, which has great inherent value. Therefore, Table 2 does not have these 3 prompt elements). Operational spacecraft's information also includes its operational status (normal or malfunction). Target information covers aspects like target types, quantity of targets, and functional categories. Environmental conditions primarily include lighting and electromagnetic conditions. There are 8 types of orbital motion characteristics information. We expand the prompt elements with specific terms or sentences based on common knowledge. Table 2 presents the expanded prompt elements corpus. While the classification of prompt elements may not be exhaustive, it suffices for testing LLMs. Thus, x_{p1} , x_{p2} , x_{p3} , and x_{p4} can be represented as elements within the sets of X_p . The omitted content can be found in Table 2. It should be noted that the specific words corresponding to each category in Table 2 are merely examples. To expand the sample size, synonyms can be used in place of the original vocabulary when generating samples.

$$\begin{aligned} x_{p1} &\in \{\text{"normal"}, \text{"malfunction"}\} \\ x_{p2} &\in \{\text{"cooperation target"}, \text{"meteor bodies"}, \dots, \text{"one"}, \text{"multiple"}, \dots\} \\ x_{p3} &\in \{\text{"full sunlight"}, \text{"backlight"}, \text{"optimal"}, \text{"irregular"}\} \\ x_{p4} &\in \{\text{"The target continuously approaches operational spacecraft, then gradually recedes after reaching the closest point."}, \dots\} \end{aligned} \quad (4)$$

Design of prompt template for interaction with LLMs

There are some tips for designing the prompt template. Liu et al. [24] conducted a comprehensive survey of the prompt method, which encompassed prompt template engineering. It involves integrating prompt elements into comprehensive LLM inputs, thereby enabling LLMs to produce the desired outputs. For spacecraft intention recognition tasks, the prompt template

Table 1. Spacecraft intentions classifications

Motion intentions	(1) Hovering	The target maintains a constant relative distance from operational spacecraft, with a relative velocity of zero.
	(2) Flyby	The relative distance between the target and operational spacecraft varies from large to small and then from small to large, yet it never becomes zero.
	(3) Flyaround	The relative distance between the target and operational spacecraft is not zero, fluctuating within a narrow range.
	(4) Rendezvous	The target has executed an orbital maneuver, resulting in a zero relative distance and a zero relative velocity with operational spacecraft.
	(5) Collision	The target's relative distance from operational spacecraft is zero, and their relative velocity is nonzero.
	(6) Retreat	The target, which originally collided with operational spacecraft, underwent orbital maneuvers and is now far away from operational spacecraft.
	(7) Randomness	The intention is unclear or time-varying.
Operation intentions	(8) Docking	During rendezvous, the target and operational spacecraft are structurally integrated through a robotic arm or docking mechanism.
	(9) Refueling	The target is typically a cooperative one, equipped with a fuel package. Operational spacecraft facilitates fuel injection.
	(10) Repair	The cooperation target encompasses maintenance tools such as robotic arms and equipment parts that operational spacecraft need to replace and add.
	(11) Grabbing	During hovering and rendezvous, the target attaches to operational spacecraft identity through robotic arms, flying claws, etc.
	(12) Photograph	The relative distance is sufficiently close for the target to photograph operational spacecraft via the camera.
	(13) Communication	The target is equipped with communication devices and emits electromagnetic signals to operational spacecraft.
Task intentions	(14) Detection	The target is noncooperative. Motion intentions are hover and flyaround. Operation intentions are photograph and gather information on operational spacecraft.
	(15) Surveillance	The target is cooperative. Both motion intentions and operation intentions are identical to detection.
	(16) Interference	The target is noncooperative. Motion intentions are hover, rendezvous, and flyaround. Operation intentions are grab and electromagnetic interference, etc. For a period of time, operational spacecraft lost the capability to perform missions in orbit.
	(17) Negotiation	Motion intentions are hover and flyaround. Operation intention is communicate.
	(18) Observation	Motion intentions is similar to that of detection, but there is no operation intentions. The target is waiting to receive instructions for operation.
	(19) Assistance	The target is cooperative. Motion intention is rendezvous. Operation intention is repair.
	(20) Deterrence	The target is noncooperative. Motion intentions are hover, flyby, rendezvous, and flyaround. Target repeatedly exhibits the aforementioned intentions. Operation intentions are uncertain.
	(21) Support	The cooperative target is to safeguard operational spacecraft. Motion intentions are hover and flyaround, but there is no operation intentions.
	(22) Experiment	The cooperative target carries experimental equipment, and motion intentions are Hover, Rendezvous, and Flyaround, with uncertain operation intentions.
	(23) Attack	The target is noncooperative. Motion intention is collide. There is no operation intentions.

engineering offers us 2 key insights. First, prefix prompts are preferred due to the directionality of LLM word prediction, making prefix prompts more suitable for LLMs. Second, manually crafted templates prove more effective than automated generation for domain-specific tasks. Shieh [25] provided some

general tips to ensure achieving the expected results. First, the template for prompts should progress from simple to complex. Second, employing instructions in prompts results in better outcomes. Third, the content of prompts should ensure that the task is specific.

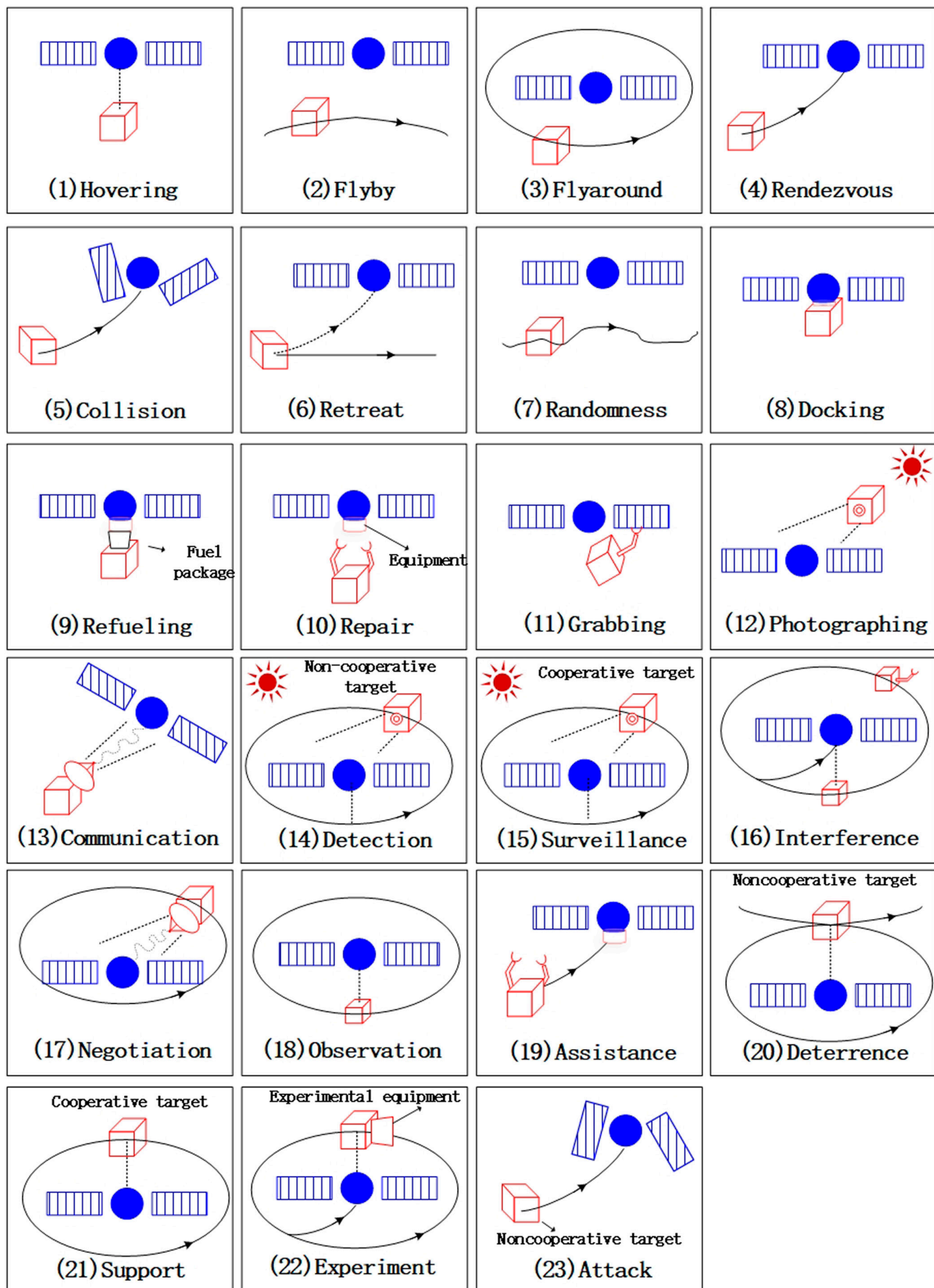


Fig. 2. Graphical illustration of spacecraft typical intentions.

Table 2. Corpus of prompt elements for spacecraft intentions recognition

Operational spacecraft's information	Target information			Environment conditions		Orbital motion characteristics
	Operation status	Target types	Quantity of targets	Functional categories	Lighting conditions	Electromagnetic conditions
Normal	Cooperation target	One	Small satellite	Full sunlight	Optimal	The target continuously approaches operational spacecraft, then gradually recedes after reaching the closest point.
						The target's relative speed with operational spacecraft is zero, yet the relative distance is not zero.
						The target maintains a relatively close and consistent distance from operational spacecraft, facilitating long-duration observations from various angles.
Malfunction	Meteor bodies, space debris	Multiple	Large satellite	Backlight	Irregular	The target was on a collision course with operational spacecraft, but executed a maneuver to avoid the collision.
						The target's orbit is chaotic and relatively close to operational spacecraft.
						The target's orbit is chaotic and relatively distant from operational spacecraft.
	Fault out of control satellite	Noncooperative satellite operating normally	Communication antenna	Operating robotic arm		The target approaches to within a zero relative distance and velocity from operational spacecraft, enabling prolonged close-range operations.
						The target approaches to within a zero relative distance and velocity from operational spacecraft, enabling prolonged close-range operations.
						The target's orbit is chaotic and relatively distant from operational spacecraft.
			Reconnaissance camera			
			Scientific experimental payload			

Similarly, the construction of the prompt template X_t is also a part of the input X . Based on above guidelines, combined with the classification of prompt elements, we initially constructed the following prompt interaction statement templates for spacecraft intention recognition tasks:

- Standard statement of operational spacecraft's information x_{12} : Operational spacecraft is ***, in *** Earth orbit, possessing ***, and operation current state is ***.
- Standard statement of target information x_{13} : The target is *** ***, spacecraft, and equipped with ***.
- Standard statement of environmental conditions x_{14} : The lighting condition is ***, and the electromagnetic condition is ***.
- Standard statement of relative orbital motion characteristics x_{15} : Based on the observed data, ***.

The instruction standard prompt statement can vary depending on the task requirements. For basic prompt testing, the instruction prompt statement x_{11} is: Based on the following prompts, output the possible target intentions. Do not include any extraneous words in the output.

$$X_t = \{x_{t1}, x_{t2}, x_{t3}, x_{t4}, x_{t5}\} \quad (5)$$

Readers can design their own templates as long as they adhere to the above principles. We used space station, low, and of significant value, filling in standard statement of operational spacecraft's information. By integrating these various templates, the final prompt template format input into the LLM is:

```
{ "Based on the following prompts, output the possible target intentions. Do not include any extraneous words in the output. ### Operational spacecraft is a space station in low Earth orbit, possessing significant value, and operation current state is ***. The target is *** ***, spacecraft, and equipped with ***. The lighting condition is ***, and the electromagnetic condition is ***. Based on the observed data, ***." }
```

The answer output is placed at the end of the prompt, conforming to the requirements of prefix prompts in prompt template engineering. The first sentence serves as a directive, specifying that the output should be "target spacecraft intention" and that only possible intentions should be provided without any additional unnecessary words. The instruction is placed at the beginning of the prompt, separated from the content by the "###" symbol, in line with the recommendation for using instructions in prompt engineering. The final directive prompt statement instructs the model to provide the answer to the prompt, aligning with the principle of specifying what the model should do rather than what it should not do. The addition of "without any additional unnecessary words" at the end is necessary to ensure that the model's output is clear and relevant, as without this guidance, the output might be fragmented or include extraneous information, which would hinder accurate analysis of the results. The standard statements are grammatically correct and chosen from the 4 types of prompt elements to provide a comprehensive set of prompt information, adhering to the principle of task specificity in prompt engineering.

These prompt interaction templates were developed at the initial stage of the task and may undergo specific modifications for fine-tuning and other subsequent processes, which will be detailed in the subsequent sections of the implementation process.

Construction of LLM test samples

Initially, we incorporate the prompt elements into the interactive prompt template to compose the input X for LLMs. Note that we used synonyms of the prompt elements in Table 2 to expand the test sample. Subsequently, we manually ascertain the spacecraft intentions that aligns with the input scenario to establish the expected output Y_e . By integrating X and Y_e , we complete the creation of LLM test samples.

Since some combinations may be unreasonable, these are manually removed. This scenario exemplifies a logical contradiction, where cooperative target information (implying coordinated behavior) is incongruously paired with collision-indicative orbital motion characteristics. Such combinations are manually deleted. In the end, we select and construct 50,688 test data sets. Its format is as follows:

```
{ "test_number": 1,
  "prompt": ["Based on the following prompts, output the possible target intentions. Do not include any extraneous words in the output. ### Operational spacecraft is a space station in low Earth orbit, possessing significant value, and operation current state is normal. The target is a large cooperative spacecraft, and equipped with operating robotic arm. Lighting condition is full sunlight, and the electromagnetic condition is optimal. Based on the observed data, the target continuously approaches us, reaches the closest distance, and then gradually moves away."],
  "answer": ["flyby, refuel."] }
```

In this format, "test_number" represents the test sample number. "prompt" is the input text X , which consists of 2 parts: instructions and the content used to fill prompt elements into the template. "answer" is the expected output Y_e , providing manually determined intent results. The "prompt" may vary slightly due to different test situations. Fine-tuning training and test data will be modified based on this format. The specific data format will be explained before each test in the Results section.

For the input content, the combination method can be represented by the formula:

$$X = \{x_{t1}, (x_{t2}, x_{p1}), (x_{t3}, x_{p2}), (x_{t4}, x_{p3}), (x_{t5}, x_{p4})\} \quad (6)$$

where (x_{t2}, x_{p1}) represents the fusion of prompt elements from operational spacecraft's information with the prompt template, and similar combinations apply for the rest.

Intention recognition improvement by fine-tuning

In this experiment, basic LLMs performed poorly in testing. Given the demanding inference needs of spacecraft intention recognition tasks and the intricacies of developing specialized knowledge bases and software, this paper employs fine-tuning techniques to enhance model performance [26]. Specifically, there are 2 techniques: P-tuning V2 and LoRA fine-tuning.

The role of fine-tuning in LLMs

First, this section offers a brief overview of the LLM training process. Fig. 3 illustrates the complete training process for LLMs applied to the spacecraft intention recognition tasks. Pretraining serves as the foundational phase in the learning process of LLMs, where the original network parameters are trained using an extensive collection of unlabeled data. This training enables the model to acquire fundamental statistical patterns and a broad understanding of language. Pretraining employs several methods, including masked language modeling (MLM), next word prediction (NWP), next sentence prediction (NSP), and general language model (GLM). In the BERT (bidirectional encoder representations from transformers) model, the MLM approach is used, where certain words in the input

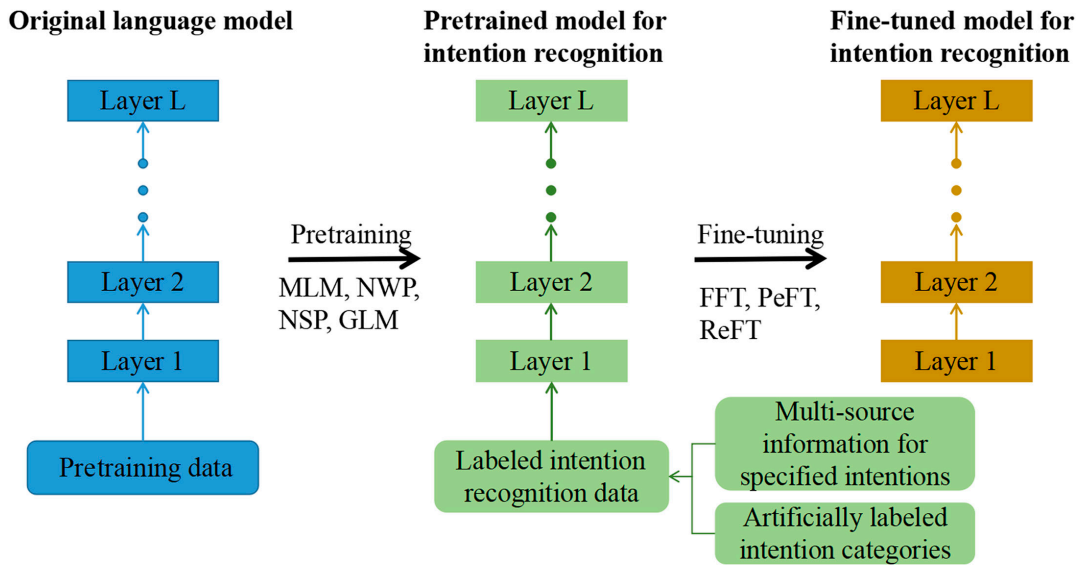


Fig. 3. LLM training process for spacecraft intention recognition.

text are randomly masked, and the model is trained to predict these obscured words. This aids in the model's learning to understand and infer context. In the GPT (generative pre-trained transformer) series models, the NWP method is employed, training the model to predict the next word following a sequence of words. This method fosters the model's ability to learn language generation. The ChatGLM model used in the paper employs a general pretraining framework based on autoregressive blank-filling objectives, known as GLM. Although this pretrained model may not exhibit exceptional performance within a specific domain, it demonstrates remarkable proficiency in terms of generalization.

Fine-tuning involves taking a pretrained model and further training it with a limited set of labeled data tailored for a particular downstream task. The dataset used for fine-tuning is considerably smaller than that used for pretraining but is specifically labeled for the task. For the purpose of this paper, the training data consist of prompts composed of multi-source information and artificially labeled intentions. The general methods of fine-tuning include full parameter tuning (FFT), parameter-efficient fine-tuning (PeFT), and representation fine-tuning (ReFT). The P-tuning V2 and LoRA-tuning methods used in this paper both belong to the category of PeFT. In this process, additional fully connected layers are often appended to the pretrained model, and both these new layers and certain existing layers of the model are then trained with the labeled data. P-tuning is an example of this approach. Alternatively, some methods involve training a reduced version of the model with a substantially lower number of parameters, as seen in LoRA-tuning. LoRA operates on the premise that updates to the pretrained model's parameters can be effectively approximated by a low-rank matrix. The goal of fine-tuning is to optimize the model's performance for the specific task, thereby enhancing its overall capabilities.

Whether it is pretraining or fine-tuning, their aim is to make the model's output $\mathbf{Y}$ closer to the desired output $\mathbf{Y}_e$. This can be expressed mathematically as:

$$L(\theta) = \sum_{i=1}^N \ell(\mathbf{Y}_i, \mathbf{Y}_{ei}) \quad (7)$$

Among them, θ signifies the model parameters, ℓ denotes the loss function, and i refers to the i th sample.

The ChatGLM2-6B and ChatGLM3-6B models selected in this paper can be used as pretrained models for further fine-tuning. Fine-tuning can be categorized into full fine-tuning and partial fine-tuning. Full fine-tuning adjusts the parameters of every layer in the pretrained model, while partial fine-tuning typically adjusts the parameters of the output layer or a few specific layers. The former requires more memory and is time-consuming. Therefore, the 2 fine-tuning methods used in this article adopt partial fine-tuning.

Principle of P-tuning V2

Prompt text $\mathbf{X}$ inputs into the first layer network of LLMs through 2 steps: tokenization and embedding. Tokenization is the process of segmenting the input text into discrete units known as tokens [22]. Each token is represented by a symbol e , with the vocabulary denoted as V , where $e \in V$. For the prompt text $\mathbf{X}$, the tokenizer generates a token sequence $\mathbf{E}$, denoted as:

$$\mathbf{X} \xrightarrow{\text{Tokenizer}} \mathbf{E} \\ \{x_{t1}, (x_{t2}, x_{p1}), (x_{t3}, x_{p2}), (x_{t4}, x_{p3}), (x_{t5}, x_{p4})\} \xrightarrow{\text{Tokenizer}} \{e_1, e_2, \dots, e_n\} \quad (8)$$

To facilitate the computer's understanding of semantics, embedding essentially maps the vocabulary V into a lower-dimensional vector space $\mathbf{H}$, where semantically similar words are positioned close to each other [27]. Token e is mapped to an element h in the vector space $\mathbf{H}$ such that $h \in \mathbf{H}$. Consequently, the token sequence is transformed into a sequence of vectors, which is then fed into the neural network as follows:

$$\{e_1, e_2, \dots, e_n\} \xrightarrow{\text{Encoder}} \{h_1, h_2, \dots, h_n\} \quad (9)$$

The value of n varies with the input $\mathbf{X}$. For standard prompts, x_{t1} comprises 25 tokens, x_{t2} 20 tokens, x_{t3} 9 tokens, x_{t4} 11 tokens, and x_{t5} 7 tokens. x_{p1} consists of 1 token, x_{p2} can be 5, 6, 7, 8, or 9 tokens, x_{p3} can be 2 or 3 tokens, and x_{p4} can be 12, 15, 18, 19, or 21 tokens, with $n \in [90, 104]$. For instruction prompts, x_{t1} is

78 tokens, setting $n \in [143, 157]$. For CoT prompts, x_{t1} is 23 tokens, leading to $n \in [88, 102]$.

Next, we explain the P-tuning V2 fine-tuning method used by the ChatGLM2-6B model [28]. Fig. 4B illustrates the principle of this fine-tuning method. In the figure, the orange boxes represent the token vectors h after tokenization and embedding the input X , the blue boxes denote the network parameters Φ_0 of the pretraining model, and the green boxes represent the newly constructed network $\Delta\Phi$. Only the parameters in the green boxes are adjustable.

This method sets up a virtual prompt part in green before the input X , referred to as soft prompts, which the github source code has set to 128 tokens. Additionally, it constructs an equal number of neurons corresponding to the embedding layer of soft prompt tokens in each layer of the LLM (the green box at the topmost layer in Fig. 4B). These new parameters of embedding layer and other layers are optimized using function 7.

Finally, the original model parameters Φ_0 are combined with the newly trained parameters $\Delta\Phi$ to obtain the fine-tuning model parameters Φ , expressed as:

$$\Phi = \Phi_0 + \Delta\Phi \quad (10)$$

In this process, the soft prompt tokens are denoted as S_p and each input can be described by the template T_{input} :

$$T_{\text{input}} = \{S_1, S_2, \dots, S_{128}, h_1, h_2, \dots, h_n\} \quad (11)$$

Principle of LoRA fine-tuning

The LoRA fine-tuning method [29] constructs 2 new low-rank network models, which are trained and then combined with the original model to create the fine-tuned model. Fig. 4C illustrates the general principle of this method.

The key difference between the LoRA fine-tuning and P-tuning V2 methods lies in the approach to constructing $\Delta\Phi$, as indicated in Eq. (10). P-tuning V2 involves the direct creation of new network parameters within the LLMs' existing parameter structure. On the other hand, LoRA fine-tuning constructs $\Delta\Phi$ externally to the LLMs, which requires that $\Delta\Phi$ and Φ share the same dimensions. LoRA fine-tuning constructs for $\Delta\Phi(\Theta)$

$$\Delta\Phi(\Theta) = AB \quad (12)$$

As shown in Fig. 4C, if $\Phi_0 \in \mathbb{R}^{d \times k}$, $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times k}$ are low-rank matrices. The forward propagation process of the model becomes

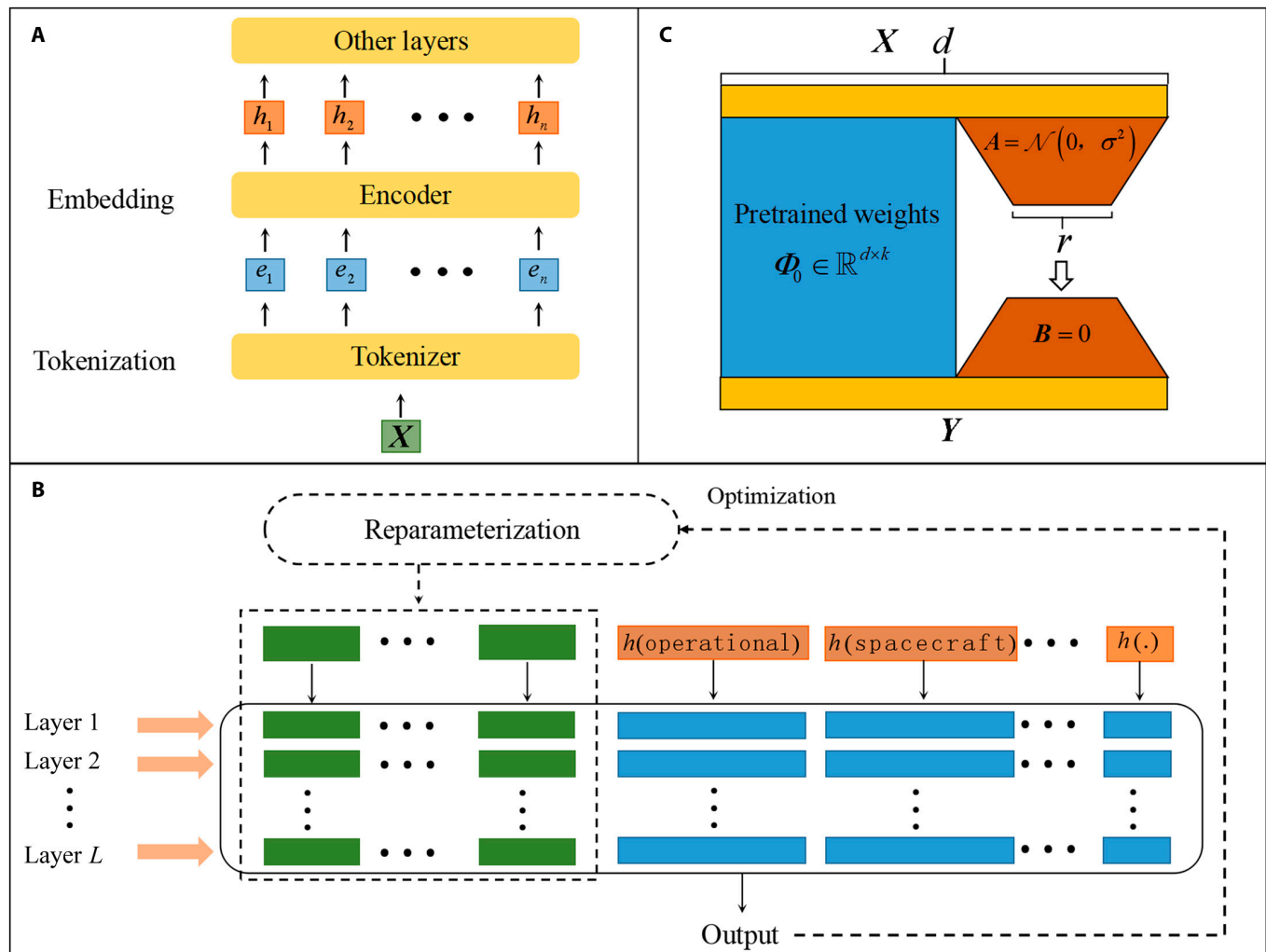


Fig. 4. (A) The input text X in LLMs first goes through 2 layers of tokenization and embedding. (B) P-tuning V2 method achieves fine-tuning of LLMs by constructing and training a network prior to input. (C) LoRA method accomplishes fine-tuning by constructing and training 2 low-rank matrices, A and B .

$$Y = X(\Phi_0 + \Delta\Phi(\Theta)) = X\Phi_0 + XAB \quad (13)$$

Since $r \ll \min(d, k)$, and A and B are low-rank matrices, the number of parameters to be trained is significantly reduced, typically accounting for only 0.1% to 3% of the original model parameters. In this experiment, $r = 8$ specified by the github source code.

Results

Experimental setup

The 2 open-source LLMs employed in the study were derived from the foundational GLM-4 architecture. However, GLM-4 lacks the capability for offline deployment, rendering the fine-tuning process cumbersome and the acquisition cost-prohibitive. The objective of the research is geared toward the automatic assessment and inference of the intent of the target spacecraft during relative orbital maneuvers, necessitating rapid responses from the LLM while ensuring safety and confidentiality. The ChatGLM2-6B and ChatGLM3-6B offline LLMs are capable of satisfying these demands.

To evaluate the impact of different prompt conditions on the model output, we used test data from 3 different prompt conditions (basic prompt, instruction prompt, and CoT prompt)

as shown in Fig. 5. The basic prompt includes only the question and necessary known information, while the instruction prompt adds all possible answers to the basic method, with the instruction part guiding LLMs to identify the appropriate answer among all possibilities. As for the CoT prompt, according to the general composition of CoT mentioned in Introduction, the instruction part of this example requires LLMs to output potential intents and provide the reasoning process. The rationale part presents the steps of human thought concluding through external knowledge and known information in the “output”. The exemplar part offers a set of “input” and “output” combinations as few-shot examples, which include the question, reasoning process, and answer.

In these formats, “content” is the input to the model, and “summary” contains the expected intention keywords. The instruction prompt data format is based on the Alpaca_data_zh_51k Chinese instruction prompt fine-tuning dataset [30]. The CoT prompt data format follows the CoT examples mentioned in the paper by Wei et al. [19], incorporating similar CoT processes into the input to teach the model to produce analogous outputs. The criterion for determining whether the model output is correct is that the output must contain any keywords (tokens) in “summary”.

All tests were conducted on a Linux server equipped with an NVIDIA A40 GPU. The graphics card with a total of 48 GB

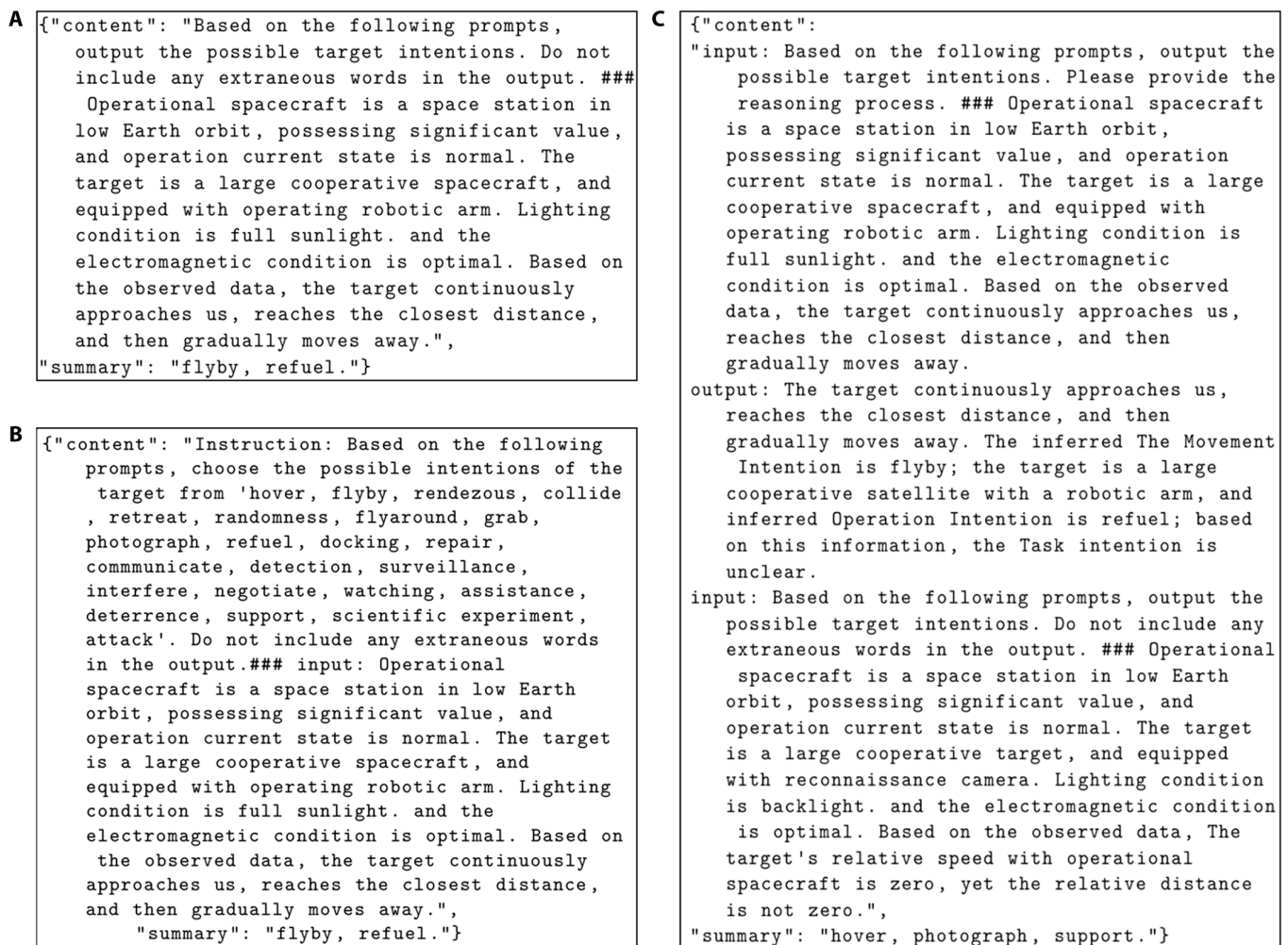


Fig. 5. The 3 prompt formats are (A) basic prompt, (B) instruction prompt, and (C) CoT prompt.

of video memory was used for training this model. Through memory resource monitoring, it was observed that the training process utilized less than 80% of the memory, while the testing process occupied approximately 20%. Otherwise, deploying the model directly requires more than 10 GB of video memory. It is important to note that the base model files for ChatGLM2 and ChatGLM3 are 12 GB in size, necessitating more than 12 GB of disk memory. Since the P-tuning method directly alters the input layer parameters, and the LoRA-tuning method retrains a new, compressed, low-rank model, the latter requires the integration of the compressed model with the base model for deployment. Under the following training parameters, training a model with the P-tuning V2 method takes approximately 4 h, while the LoRA-tuning method requires less than 3 h. During the testing phase, different models under various prompt conditions exhibit varying test times, as do different samples. Among the 6 models tested, the basic and instruction prompts have test times on the order of seconds, while the CoT prompts have test times on the order of 10 s. Due to the longer text sequences generated by CoT prompts, the model's prediction output time is longer.

ChatGLM2-6B series experimental testing

ChatGLM2-6B basic model experimental testing

The related work on the ChatGLM2-6B model refers to the research by Zeng et al. [31]. Results of the basic prompt tests for the ChatGLM2-6B base model are shown in Table 3. Examination of the test results demonstrated semantic congruence, as exemplified by test data number 73 (refer to Table 3), where the model's response "Retreating collisions" exhibited a strong semantic correlation with the anticipated output "Retreat". This observation underscores the ChatGLM2-6B base model's potential in the realm of spacecraft intention recognition. Furthermore, Fig. 6A illustrates the accuracy metrics of the ChatGLM2-6B base model under varying prompt conditions. The data clearly indicate that the prompt conditions have a pronounced impact on accuracy, with the instruction prompt yielding the most favorable outcomes. Nonetheless, the aggregate accuracy across the 3 experimental conditions remains modest, implying that the base model's training parameters may be insufficient for effectively addressing intricate reasoning challenges.

ChatGLM2-6B P-tuning V2 model experimental testing

To address the issue of low accuracy mentioned above, we conducted fine-tuning on the base model, specifically using the ChatGLM2-6B model with the P-tuning V2 method. Fine-tuning initially involves constructing task-specific datasets.

These datasets are used to train the model for the purpose of fine-tuning. For the 2 types of fine-tuning models described below, we randomly allocated 10,000 samples from the dataset as the training set. From the remaining 40,688 samples, an additional 10,000 were selected to constitute the testing set. The training parameters were set as follows: 1,500 iterations, 128 prompt tokens, learning rate of 0.02, max token input length of 256, max token output length of 128, and quantization level int8.

Instruction prompt fine-tuning model

To address the issue of low semantic accuracy, we presented all potential spacecraft intention vocabularies in the prompts, allowing the LLMs to infer possible intents based on instructions. Since the training format of the ChatGLM2-6B model consists of standard input–output pairs, both the training and testing sets are formatted accordingly.

Using the P-tuning V2 fine-tuning method, we trained the ChatGLM2-6B model with instruction prompt data. Table 4 presents partial results of the ChatGLM2-6B–instruction prompt–P-tuning model under instruction prompt conditions. The results demonstrate that the LLM outputs indeed reflect intention vocabulary, indicating the effectiveness of fine-tuning training in addressing semantic similarity issues. Fig. 6B illustrates that the model trained with instruction prompt data achieves the highest accuracy in instruction prompt testing, suggesting better adaptation to the training data format. Meanwhile, compared to Fig. 6A, the accuracy of the fine-tuned model has consistently improved under all 3 prompt conditions, indicating enhanced generalization of the fine-tuned model for spacecraft intention recognition.

CoT prompt fine-tuning model

Given the low accuracy of the previous fine-tuning, and recognizing that CoT prompts can significantly enhance the model's reasoning capabilities, we opted to fine-tune the model using CoT prompts. The training data included the aforementioned CoT prompt test data.

Table 5 displays partial results of the ChatGLM2-6B–CoT prompt–P-tuning model under CoT prompt conditions. The fine-tuned model output indeed imitates the style of the thought chain, providing a complete thought chain process and accurate results. Fig. 6C shows that the model's accuracy in CoT prompt testing is highest (99.81%) after fine-tuning with CoT prompts. We have successfully realized our aim of employing CoT to enhance the model's performance across intricate reasoning challenges. Simultaneously, the test results of the other 2 prompts exhibited substantial advancement relative to

Table 3. ChatGLM2-6B basic model test results

Test number	Correct keywords corresponding to the labeled intention	LLM output	True or false
73	Retreat, photograph, support	Avoiding collisions	False
74	collision, absence of operation intentions, absence of task intentions	The target underwent a collision with our space station and generate space debris.	True
75	Rendezvous, photograph, detection	The possible intention of the target is to carry out spacecraft maintenance and operations.	False

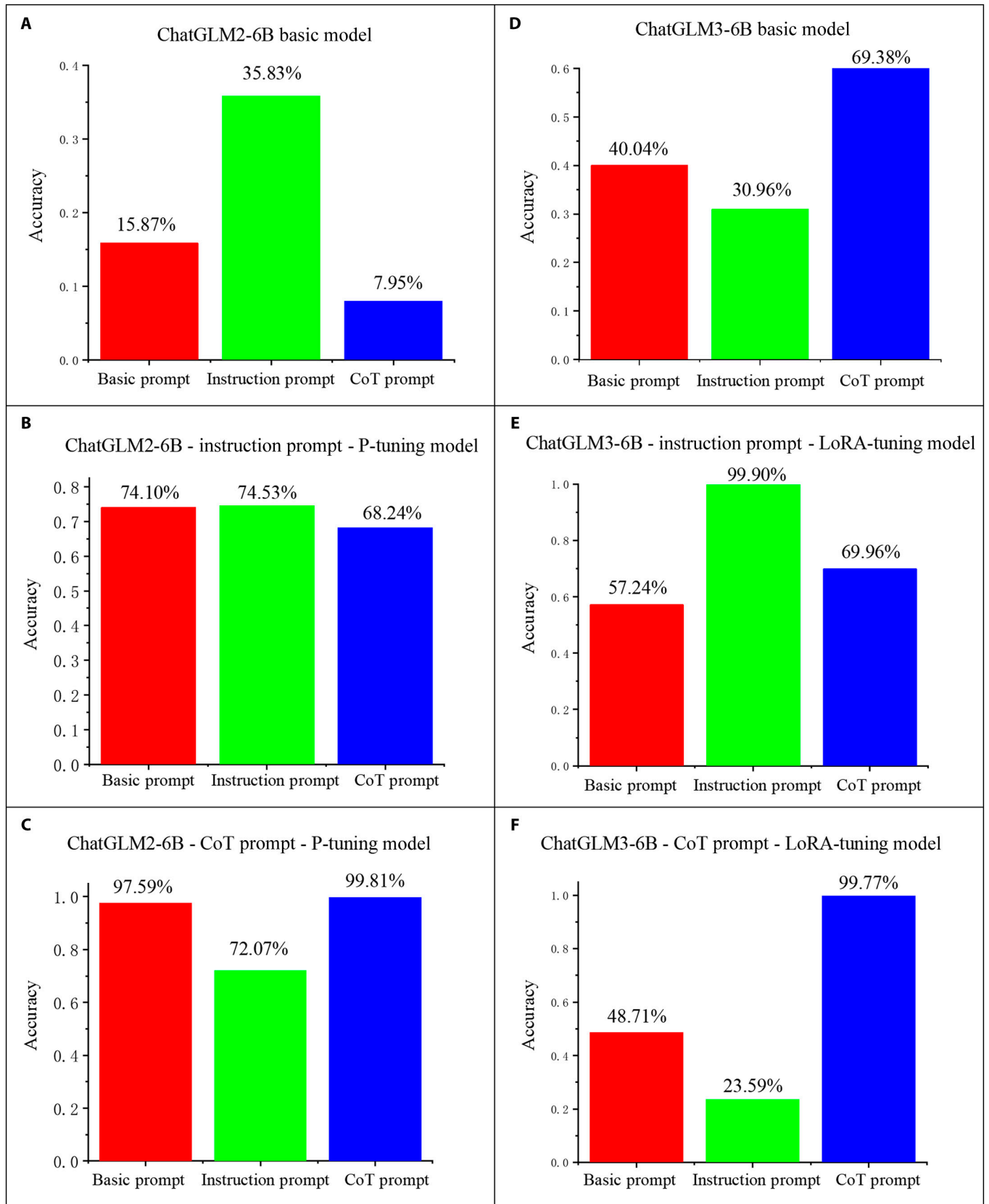


Fig. 6. The testing accuracy of 3 different prompts on 6 different model are (A) ChatGLM2-6B basic model, (B) ChatGLM2-6B-instruction prompt-P-tuning model, (C) ChatGLM2-6B-CoT prompt-P-tuning model, (D) ChatGLM3-6B basic model, (E) ChatGLM3-6B-instruction prompt-LoRA-tuning model, and (F) ChatGLM3-6B-CoT prompt-LoRA-tuning model.

Table 4. ChatGLM2-6B–instruction prompt–P-tuning model test results

Test number	Correct keywords corresponding to the labeled intention	LLM output	True or false
1303	Flyby, grabbing, support	Absence of operational intentions, absence of task intentions	False
1304	Communication, photography, support	Flyby, communication, negotiation	True
1305	Absence of operation intentions, support	Randomness, absence of operation intentions, absence of task intentions	True

Table 5. ChatGLM2-6B–CoT prompt–P-tuning model test results

Test number	Correct keywords corresponding to the labeled intention	LLM output	True or false
58	Rendezvous, communication, support	The relative distance and velocity between the target and our spacecraft eventually become zero, allowing for close range and long-term operations on our spacecraft. Inferring that the intention of orbital motion is rendezvous; The target is a small satellite, and the inferred operational intention is unclear; The electromagnetic conditions are good, and the target is sending a signal to us, inferring that the mission intention may be negotiation.	True
59	Randomness, absence of operation intentions, absence of task intentions	The target orbital motion has no regularity, and it is inferred that the intention of orbital motion is characterized by randomness. The target is relatively far away from our side, so it is inferred that the threat is relatively small. Owing to the satellite's malfunction, it is deduced that there is an absence of operation intentions or task intentions.	True

the ChatGLM2-6B baseline (Fig. 6A), indicative of an enhanced generalization ability within the fine-tuned model.

ChatGLM3-6B series experimental testing

ChatGLM3-6B basic model experimental testing

The outcomes of the aforementioned tests distinctly show variability in the accuracy across different prompts. To assess the efficacy of the CoT approach introduced by Wei et al. [19] and the instruction prompt framework proposed by Shieh [25] across various models, the 3 prompts employed in ChatGLM2 were additionally evaluated using the ChatGLM3-6B model [32].

Partial results of the ChatGLM3-6B basic model under basic prompts are shown in Table 6. The table indicates that the model outputs complete sentences and categorizes different intents, demonstrating a more human-like thought process compared to ChatGLM2-6B. The accuracy of the ChatGLM3-6B basic model under 3 different prompt conditions is shown in Fig. 6D. For the ChatGLM3-6B basic model, CoT prompts yielded better results. Compared to the ChatGLM2-6B basic model, this model showed improved accuracy in both basic

and CoT prompt tests, indicating that the base performance of ChatGLM3-6B is indeed superior to that of ChatGLM2-6B. Additionally, the experimental results illustrate that the capabilities of different language models can vary significantly under identical task conditions. This variation is related to the training data and training methods of the language models.

ChatGLM3-6B LoRA fine-tuning model experimental testing

The previous instruction-based fine-tuning results for the ChatGLM2-6B model were suboptimal, largely due to the selection of the instruction prompt fine-tuning data format. The alpaca_data_zh_51k Chinese instruction prompt fine-tuning dataset includes 3 components: instruction, input, and output. This dataset comes in 2 formats, one of which includes input content, while the other does not. Research by Cui et al. [30] indicated that training with data containing input content yields better results. To align with the instruction prompt fine-tuning data format that includes content for all 3 components (instruction, input, and output), we selected ChatGLM3-6B [33] as the base model.

Table 6. ChatGLM3-6B basic model test results

Test number	Correct keywords corresponding to the labeled intention	LLM output	True or false
180	Hovering, absence of operation intentions, <u>experiment</u>	The intention of the target may be to conduct scientific <u>experiment</u> .	True
181	Hovering, absence of operation intentions, support	Possible intentions of the target: observation, prediction, communication, localization, identification, tracking, and communication.	False

The model fine-tuning employs the LoRA method. The multi-turn dialogue fine-tuning data template for ChatGLM3-6B is as follows:

```
{
  "conversations": [
    {
      "role": "system",
      "content": "<system prompt text>"
    },
    {
      "role": "user",
      "content": "<user prompt text>"
    },
    {
      "role": "assistant",
      "content": "<assistant response text>"
    }
  ]
}
```

This multi-turn dialogue fine-tuning data include 3 roles: system, user, and assistant, which correspond directly to the instruction prompt fine-tuning components: instruction, input, and output. The system role is used for the instruction label, the user for the input label, and the assistant for the output label.

Both the training and validation datasets contain 10,000 samples each. The training parameters are set with a learning rate of 0.00005, a maximum of 3,000 iterations, max token input length of 256, max token output length of 128, a LoRA rank of 8, and a scaling factor of 32.

Instruction prompt fine-tuning model

The training and validation data format for instruction prompt fine-tuning is shown in Fig. 7A.

Table 7 presents the outcomes of the ChatGLM3-6B model's instruction prompt-based LoRA-tuning, demonstrating a high congruence between the LLM's output and the intended objectives. As illustrated in Fig. 6E, the model achieves an accuracy of 99.90%, which is the highest observed across all tested models. The fine-tuning of instruction prompts on the ChatGLM3 base model has proven to be highly effective. The result emphasizes the utility of the instruction fine-tuning approach for the purpose of discerning the intentions of the target spacecraft. Nonetheless, in contrast to the findings in Fig. 6D, the enhancements for the remaining 2 prompts are negligible, suggesting that the fine-tuning model exhibits limited generalizability.

CoT prompt fine-tuning model

Similarly, the training and validation data format for CoT prompt fine-tuning is shown in Fig. 7B.

Table 8 presents partial results of the ChatGLM3-6B CoT prompt LoRA-tuning model under CoT prompt conditions. The output of LLMs provides a complete thought process and accurate results. As depicted in Fig. 6F, the model exhibits an average accuracy of 99.77% under CoT prompt conditions, signifying robust adaptation to the training dataset. The CoT prompt demonstrates considerable efficacy in fine-tuning the ChatGLM3 base model. Conversely, the performance under instruction prompts registers a decline relative to the ChatGLM3-6B baseline

(Fig. 6D), suggesting a reduction in the model's generalization ability.

Experimental testing of LLM robustness for spacecraft intention recognition

The preceding discussion involved conducting basic tests on 6 models under 3 types of prompts. However, real-world input formats are unlikely to conform strictly to the aforementioned 3 prompt types. Consequently, it is imperative to evaluate the generalization capabilities of the models. To better evaluate model performance, we refer to the holistic evaluation of language models (HELM) method proposed by Liang et al. [34]. HELM mentions several evaluation attributes, including accuracy, calibration, generalization ability, adaptation ability, robustness, efficiency, bias and stereotypes, fairness, and toxicity. Although we cannot comprehensively conduct all these evaluations, we have focused on accuracy and robustness testing. Accuracy testing includes various metrics such as precision, recall rate, and F1 score. The paper pertains to fuzzy classification problems, where recall rate and F1 score are not computable. Here, we only calculate accuracy, which already demonstrates that the model can be successfully applied to simple spacecraft intention recognition task. Robustness emerges as the critical factor in assessing a model's generalization ability.

This section conducts robustness testing, which measures the model's resistance to perturbations or noise in the input data. One method to evaluate model robustness is to introduce perturbations to the text input and observe changes in the model's output. These perturbations can be broadly classified into 2 categories: adversarial perturbations [35–38] and non-adversarial perturbations [39]. Adversarial perturbations involve intentionally modifying the input to mislead the model into making incorrect predictions. Although these perturbations might not affect human judgment, they significantly impact the model's prediction results. Non-adversarial perturbations, on the other hand, are more natural and random changes to the input content. These perturbations are not designed to cause model errors but to simulate the complexity of real-world inputs. In this experiment, we set up 3 conditions to simulate real-world scenarios, all of which involve non-adversarial perturbations.

Condition 1: Changing the order of prompts. The order of the 4 prompt statements is altered and combined to form new robustness test samples. The expected intentions is reassessed by humans. This condition aims to simulate uncertainty in the data input sequence during genuine intention recognition.

Condition 2: Missing prompts. One or 2 of the 4 prompt statements in the test samples are randomly removed to form

A <pre>{ "conversations": [{ "role": "system", "content": "Based on the following prompts, choose the possible intentions of the target from 'hover, flyby, rendezvous, collide, retreat, randomness, flyaround, grab, photograph, refuel, docking, repair, communicate, detection, surveillance, interfere, negotiate, watching, assistance, deterrence, support, scientific experiment, attack'. Do not include any extraneous words in the output." }, { "role": "user", "content": "Operational spacecraft is a space station in low Earth orbit, possessing significant value, and operation current state is normal. The target is a large cooperative target, and equipped with operating robotic arm. Lighting condition is full sunlight. and the electromagnetic condition is optimal. Based on the observed data, the target continuously approaches us, reaches the closest distance, and then gradually moves away." }, { "role": "assistant", "content": "flyby, refuel" }] }</pre>	B <pre>{ "conversations": [{ "role": "system", "content": "Based on the following prompts, output the possible target intentions. Please provide the reasoning process." }, { "role": "user", "content": "Operational spacecraft is a space station in low Earth orbit, possessing significant value, and operation current state is normal. The target is a large cooperative target, and equipped with operating robotic arm. Lighting condition is full sunlight. and the electromagnetic condition is optimal. Based on the observed data, the target continuously approaches us, reaches the closest distance, and then gradually moves away." }, { "role": "assistant", "content": "The target continuously approaches us, reaches the closest distance, and then gradually moves away. The inferred Movement Intention is flyby; the target is a large cooperative, and equipped with a robotic arm, and inferred Operation Intention is refuel; based on this information, the Task intention is unclear." }] }</pre>
---	---

Fig. 7. The format of training and validation data (A) pertains to instruction prompts, while (B) pertains to CoT prompts.

Table 7. ChatGLM3-6B–instruction prompt–LoRA-tuning model test results

Test number	Correct keywords corresponding to the labeled intention	LLM output	True or false
6,347	<u>Retreat</u> , absence of operation intentions, <u>absence of task intentions</u>	<u>Retreat</u> , absence of operation intentions, <u>absence of task intentions</u>	True
6,348	<u>Rendezvous</u> , photograph, surveillance	<u>Rendezvous</u> , absence of operation intentions, support	True
6,349	<u>Flyby</u> , absence of operation intentions, absence of task intentions	<u>Flyby</u> , absence of operation intentions, support	True

Table 8. ChatGLM3-6B–CoT prompt–LoRA-tuning model test results

Test number	Correct keywords corresponding to the labeled intention	LLM output	True or false
67	<u>Flyby</u> , communication, negotiation	The target keeps getting closer to us, and after reaching the closest distance, it gradually moves away. This behavior suggests that the orbital trajectory is intended as <u>flyby</u> .	True
68	<u>Retreat</u> , absence of operation intentions, absence of task intentions	The target is maneuvered to retreat from potential collision with our assets. This suggests that the intention of its orbital motion is to <u>retreat</u> .	True

new test samples. The expected intentions are reassessed by humans. The goal of this condition is to simulate scenarios involving missing incoming data in real world.

Condition 3: Modifying prompt content. The content of the prompt statements is rephrased and expressed differently, ensuring that the expected intentions remain unchanged when judged by humans. The objective of this condition is to replicate the transformation of incoming data into various text language formats in real-world scenarios.

The robustness testing involves a subset of 8,448 samples, manually rewritten from a total of 50,688 to create perturbed data under 3 specified conditions. The testing framework is as follows:

Dataset: It includes the refined 8,448 samples, divided into 8,448 standard test samples for the basic prompt and an additional 8,448 perturbed samples for each of the 3 conditions. Subsequently, a random batch of 1,000 samples is drawn for testing in each round.

Baseline models: ChatGLM2 and ChatGLM3 serve as the reference models.

Test models: Four models are tested, including ChatGLM2 with P-tuning for 2 prompt types and ChatGLM3 with LoRA-tuning for 2 prompt types.

Evaluation metric: The primary measure for assessing model robustness is accuracy.

ChatGLM2-6B series model experimental testing of robustness

We conducted robustness tests on 3 models: the ChatGLM2-6B base model, ChatGLM2-6B instruction prompt P-tuning model,

and ChatGLM2-6B CoT prompt P-tuning model, using the 3 prompts with perturbation and standard prompts. The initial 3 clusters in Fig. 8 encapsulate a synopsis of the precision outcomes derived from these 12 assessments. The orange bar delineates the direct selection of 1,000 specimens from the 50,688-sample corpus for examination, yielding results that echo the corresponding cues of the model illustrated in Fig. 6. The red bar signifies alterations in the sequence of prompts, the green bar denotes the omission of prompts, and the blue bar represents the substitution of prompts with semantically analogous terms. The black dashed line charts the mean precision across the quartet of robustness evaluations for each model. Observations indicate that the basic model may exhibit higher accuracy with perturbed samples compared to undisturbed ones, as the effect of perturbations on the basic model is uncertain—whether positive or negative. Conversely, for fine-tuned models, the accuracy in testing undisturbed samples consistently surpasses that of perturbed samples, since these models are trained specifically on undisturbed data. In broad strokes, the fine-tuned model has eclipsed the baseline model in terms of precision. Table 9 lays out the standard deviations of the 4 result sets garnered from perturbation analyses for each model. A reduced standard deviation signifies a model's diminished susceptibility to perturbations, thereby reflecting its enhanced robustness and superior generalization ability. It is evident that the basic model of ChatGLM2 demonstrates enhanced robustness and a superior generalization. Employing CoT prompts in the fine-tuned model yields a standard deviation akin to the base model, suggesting only a negligible reduction in robustness.

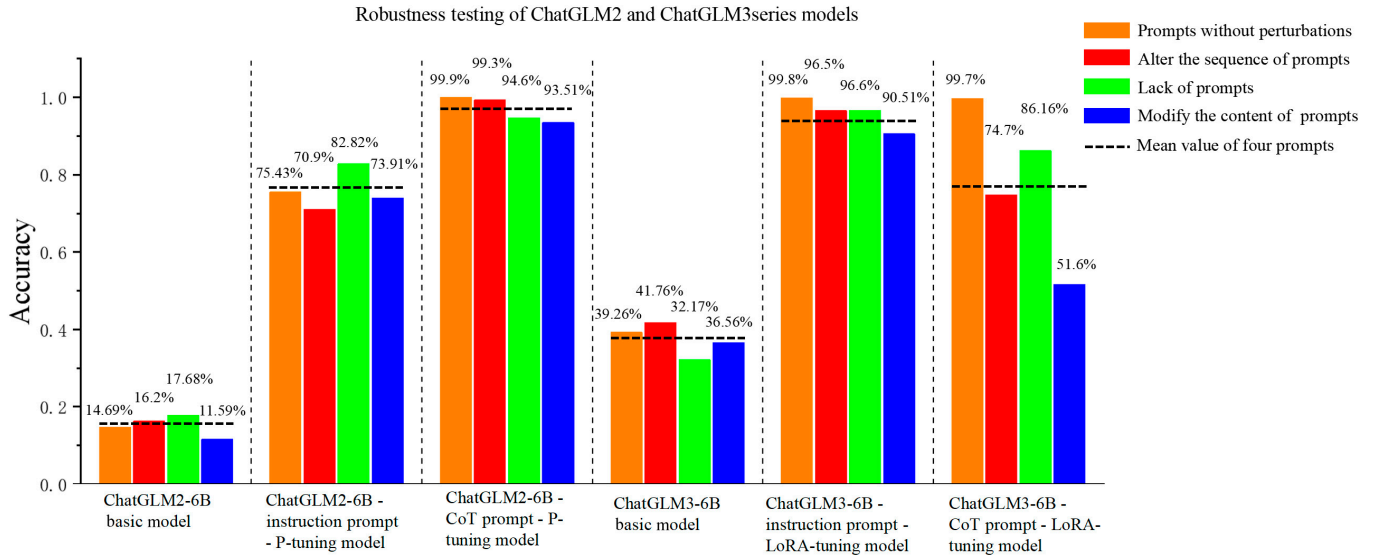


Fig. 8. The accuracy of robustness testing for the ChatGLM2 and ChatGLM3 series models across 3 categories of perturbation prompts and basic prompts.

Table 9. Standard deviation of 6 models for perturbations test

Standard deviation	Basic model	Instruction prompt fine-tuned model	CoT prompt fine-tuned model
ChatGLM2-6B	0.026038561	0.050660471	0.032414541
ChatGLM3-6B	0.041037412	0.038774165	0.203741601

ChatGLM3-6B series model experimental testing of robustness

Similarly, the last 3 sets in Fig. 8 display the ChatGLM3 series' test outcomes with and without perturbation prompts. The fine-tuned model exhibits enhanced accuracy versus the base model. The peak accuracy in undisturbed tests of the fine-tuned model also suggests strong adaptation to the training data. Table 9 horizontal comparison reveals that the ChatGLM3 model, refined with instructional prompts, attains peak robustness, outperforming the base model. Longitudinal analysis indicates that instructional prompt fine-tuning enhances the ChatGLM3 model's robustness, while CoT prompt fine-tuning optimizes the ChatGLM2 model's robustness. The effectiveness of fine-tuning methods varies with the model's architecture.

Discussion

We have discussed the following 4 points regarding accuracy testing and robustness testing.

- The basic model's capabilities significantly impact performance on specific tasks, and the same fine-tuning methods yield varying results across different base models. A comparison of the base models' accuracy in Fig. 6 indicates that the ChatGLM3-6B model exhibits higher overall performance compared to the ChatGLM2-6B. Tests conducted by Tsinghua University [32] show that the ChatGLM3-6B-Base model outperforms the ChatGLM2-6B on fundamental datasets such as GSM8K, MATH, and BBH, consistently demonstrating superiority. Experimental results from the spacecraft intention recognition task further corroborate this finding.
- Different prompting conditions can drastically affect the output of the same model. Models fine-tuned with specific prompts exhibit adaptability to those prompts. Fig. 6 reveals significant accuracy differences among the 3 prompting conditions for the same model across 18 test scenarios in this study. The highest precision was achieved under the same prompting conditions as the training set among all fine-tuned models.
- In general, basic models exhibit better robustness compared to fine-tuned models. The high standard deviation of the ChatGLM3-CoT prompt fine-tuned model in Table 9 indicates poor robustness, suggesting that while fine-tuned models are more suited to specific prompting conditions, they are more sensitive to non-adversarial data interference and less resilient. However, the P-tuning model with ChatGLM2-CoT prompts and the ChatGLM3-6B instruction prompt LoRA-tuning model performed well in accuracy and robustness tests, highlighting their suitability for the spacecraft intention recognition task.
- The CoT prompting method presents challenges in guiding the reasoning process of smaller models (less than 10B) [40], but this study offers a different perspective. Fig. 6C and F shows that the ChatGLM2-6B model fine-tuned with P-tuning V2 and the ChatGLM3-6B model fine-tuned with LoRA under CoT prompting conditions performed similarly well. Moreover, results from Tables 5 and 8 suggest that CoT prompts can elicit a reasoning process in the spacecraft intention recognition task. Liu et al. [28] note that previous rapid fine-tuning methods have not performed well on models below 10B. The

P-tuning V2 method primarily adapts to the dataset and task in testing. However, for more complex reasoning tasks, P-tuning V2 may be ineffective for smaller models, as evidenced by the CoT prompt test results.

In conclusion, we leveraged the logical reasoning capabilities of LLMs to integrate and assess multi-source information, achieving spacecraft intention recognition through precise model fine-tuning. Specifically, the study developed a spacecraft intention language model interacting with LLMs, encompassing 23 intentions in 3 categories, 7 prompting elements in 4 categories, and 50,688 test samples. Accuracy and robustness tests were conducted on the base models ChatGLM2-6B and ChatGLM3-6B, and their respective fine-tuned counterparts. For both models, fine-tuning methods increased accuracy by up to 83.94% and at least 58.66%; however, robustness relatively declined, with the standard deviation expanding by up to 4.96 times and at least 1.25 times. The models fine-tuned with different prompt formats achieved the highest accuracy when tested with data in the corresponding prompt format, indicating adaptation to the data format of the respective tasks. Notably, the ChatGLM3-6B model improved with LoRA instruction prompt data achieved an outstanding test accuracy of 99.9%, representing the most significant accuracy improvement with minimal robustness reduction, highlighting LLMs' aptitude for spacecraft intention recognition tasks. Furthermore, this study expands the scope of LLMs in the aerospace field, providing a stricter and more effective technology for identifying spacecraft intentions and laying a solid foundation for subsequent research. It offers a comprehensive definition of spacecraft intentions, setting clear theoretical benchmarks and standard references for future studies. By using P-tuning and LoRA techniques, the study significantly improved LLM accuracy in complex tasks, providing valuable guidance for future applications.

Beyond the achieved innovations, the research presented in this paper has 4 notable limitations.

- Satellite-based computational capabilities are inadequate: Given the current scenario, although satellite-transmitted data can be utilized for intent recognition using LLMs once received on the ground, the installation of high-performance computing hardware on satellites remains impractical, thus hindering autonomous recognition capabilities.
- LLMs exhibit limited reasoning: The focus of this paper is primarily on standard scenarios. When it comes to situations involving multiple objectives, collaborative operations, or complex strategies, the reasoning capabilities of LLMs are likely to be insufficient.
- Test performance is compromised: Despite the model's capability to achieve up to 99.9% accuracy, robustness tests have revealed a decline in performance across all models when exposed to perturbation test sets, indicating a degradation in real-world scenario testing.
- Interpretability remains elusive: While this paper employs CoT to demonstrate the model's reasoning process, the industry still lacks a comprehensive theoretical understanding of CoT, suggesting that the perceived logic may not always reflect actual logical processes.

Moving forward in future research, the following 2 considerations should be taken into account.

- To address the issue of fine-tuned models' poorer robustness compared to base models in robustness tests, incorporating

a small portion of non-adversarial perturbation samples into training could enhance model generalization [41], thereby improving the robustness of the trained models. Alternatively, expanding the parameter set of the base model and adopting those exceeding the 10B threshold can enhance accuracy significantly. These models have shown robustness in CoT evaluations and are adept at elucidating thought processes with limited CoT instances [19]. Consequently, this strategy minimizes the expenses associated with the manual production of extensive thought chain exemplars.

- In light of the latest developments in multimodal LLMs [42], more general cases should be considered in future applications of spacecraft intention recognition, including integrating various types of information from sensors, such as image and distance data, for comprehensive judgment. Intention recognition no longer relies solely on language text, thereby facilitating decision-making in orbit.

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Data Availability

The data are available from the authors upon a reasonable request.

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